

**ECA56 –Advanced Wireless Communication Systems / 5G / 6G
Communication Systems**

Exp: 6 5G NR Frame Structure Analysis with Different Subcarrier Spacing

Aim :

To analyze the 5G NR frame structure by studying the effect of different subcarrier spacings, slot durations, and OFDM symbols using MATLAB.

Procedure :

- Define the supported 5G NR subcarrier spacing values.
- Display the values using MATLAB.
- Plot the subcarrier spacing for different numerologies.

Objective – 1: Analyze Different Subcarrier Spacing Values

Program :

```
clc;  
clear;  
close all;  
scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)  
mu = 0:4; % Numerology Index  
bar(scs)  
grid on  
set(gca,'XTick',1:5)  
set(gca,'XTickLabel',mu)  
xlabel('Numerology Index (\mu)')  
ylabel('Subcarrier Spacing (kHz)')
```

title('5G NR Subcarrier Spacing for Different Numerologies')

Output :

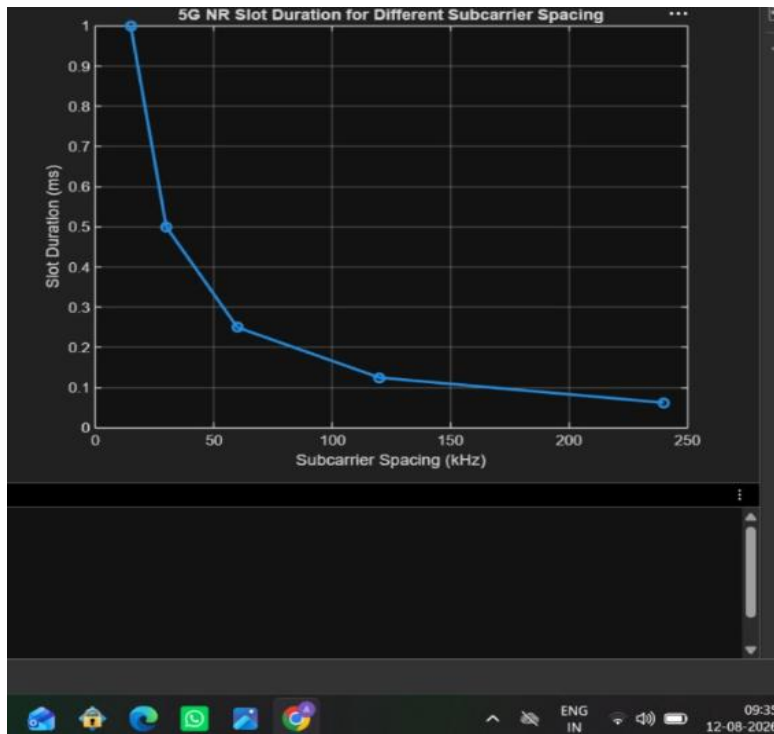


Objective – 2 : Compare Slot Duration

Program :

```
clear;
close all;
scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)
slot = [1 0.5 0.25 0.125 0.0625]; % Slot Duration (ms)
plot(scs, slot, 'o-', 'LineWidth', 2)
grid on
xlabel('Subcarrier Spacing (kHz)')
ylabel('Slot Duration (ms)')
title('5G NR Slot Duration for Different Subcarrier Spacing')
```

Output :

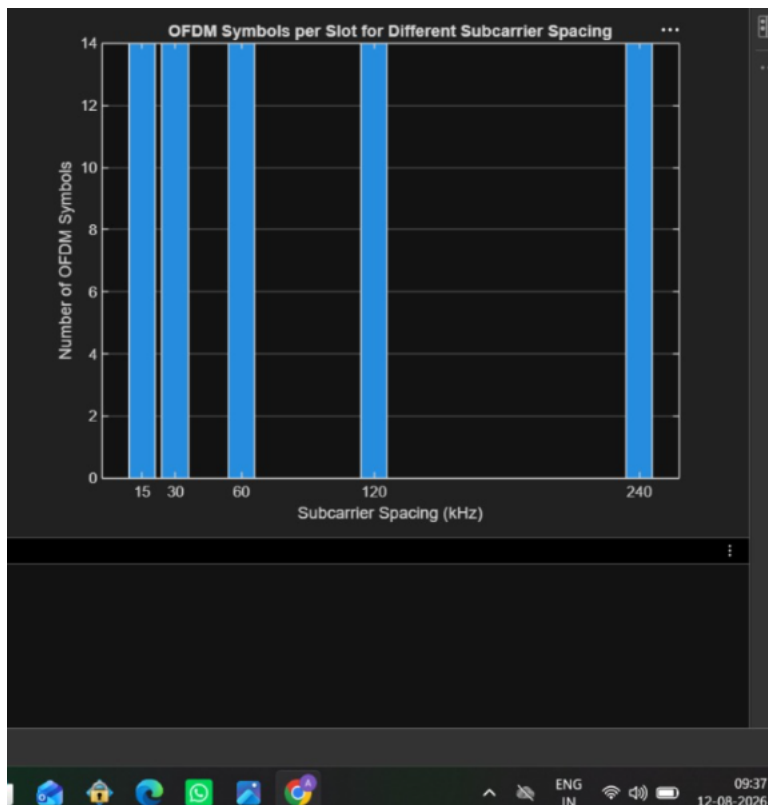


Objective – 3 : Study the Number of OFDM Symbols

Program :

```
clc;
clear;
close all;
scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)
symbols = [14 14 14 14 14]; % OFDM Symbols per Slot
figure
bar(scs, symbols)
grid on
xlabel('Subcarrier Spacing (kHz)')
ylabel('Number of OFDM Symbols')
title('OFDM Symbols per Slot for Different Subcarrier Spacing')
ylim([0 16])
```

Output :



Result:

The MATLAB analysis successfully demonstrated the flexible numerology of 5G NR. The results showed that increasing the subcarrier spacing decreases the slot duration while maintaining a constant number of OFDM symbols per slot under the normal cyclic prefix configuration.